

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 2b - Modeling Agency

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Introducing Agency

So far, we have considered the different elements that are used to model the environment, capturing its dynamics, uncertainty, etc.

We also considered **actuation** and **actions** that affect the environment, and **policies** that prescribe which actions are applied at a given state or belief.

We now consider **agency** - the ability to decide how to act, i.e., the process of computing policies.

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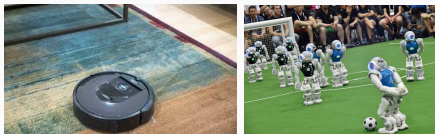
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Reminder: Reactive vs. Rational Agent



- **Reactive or Reflex agent:**
 - simple reactions to stimulus (short-term)
 - fast response times and limited resource usage
- **Rational agent:**
 - extended reasoning capabilities
 - can achieve complex objectives
 - utility maximizing
 - can learn new knowledge and react to change

We focus on **rational** agents.

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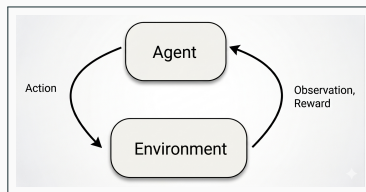
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An Agent



An agent does not have a pre-defined policy.
It has **agency** to decide how to act.

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Policies for Rational Agents

- Typically cannot be described explicitly for each state and require a compact representation.
- An **optimal policy**, denoted π^* is *better* than or equal to all other policies.
- Since our focus is on rational agents with a clearly defined utility measure, an optimal policy maximizes the agent's utility function $\mathcal{U}$.



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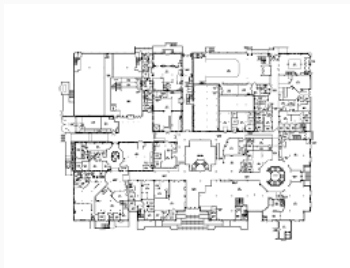
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How to come up with 'good' policies ?



Why Sequential Decision-Making?

Many real-world problems require reasoning **beyond immediate outcomes**.

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Why Sequential Decision-Making?

Many real-world problems require reasoning **beyond immediate outcomes**.

- **Information gathering affects future decisions**
 - An agent may choose to sense the environment before acting.
- **Actions influence future opportunities**
 - An agent may take a longer path now to avoid obstacles later.
- **Short-term rewards may conflict with long-term goals**
 - Energy systems may incur temporary costs to maintain grid stability.
- **Decisions accumulate over time**
 - Poor early choices can lead to irreversible states.

Sequential decision-making provides a framework for **optimizing behavior over time rather than step-by-step reactions**.

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Many Approaches to Control

Sequential decision-making in AI is not the only framework used to control dynamical systems. Many approaches have been developed in other research disciplines.

In robotics, for example, classical **feedback controllers** (e.g., PID) are designed to react to the current system state without explicitly planning far into the future.

In contrast, **optimal control** includes methods for computing a sequence of control inputs $u_0, u_1, \dots, u_T$ that minimizes a cost function over time while satisfying the system dynamics.

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Many Approaches to Control*

One widely used approach is **Model Predictive Control (MPC)**, which repeatedly solves a short-horizon optimization problem and applies only the first control action before replanning.

Example: An autonomous car approaching a curve computes a steering and acceleration plan for the next few seconds. It executes only the first control input, observes the new state of the vehicle, and then recomputes the plan.



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Sequential Decision-Making

Our focus in this course is on **sequential decision-making**, as studied under **AI and reinforcement learning (RL)**.

In these settings, an agent must choose actions **over time**, where each action affects the future state of the environment.

We will mostly assume the state space is **discrete and finite**, although it may be extremely large.

This contrasts with approaches where the system dynamics are often modeled in **continuous spaces** and a major challenge is computing or approximating good **discretizations**.

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A note on notations

When we modeled the system as a **stochastic process**:

- State is a random variable: $X_t \in \mathcal{X}$
- Focus: probabilistic evolution and inference

$$\Pr(X_{t+1} \mid X_t), \quad \Pr(O_t \mid X_t)$$

When considering **decision-making**:

- A state is typically represented as: $s_t \in \mathcal{S}$
- The transition probabilities are represented by a **transition function**:

$$\mathcal{P}(s_t, a, s_{t+1}) = \Pr(X_{t+1} = s' \mid X_t = s, A_t = a)$$

- Observation probabilities are represented by a **sensor function**:

$$\mathcal{O}(o \mid s) = \Pr(O_t = o \mid X_t = s)$$

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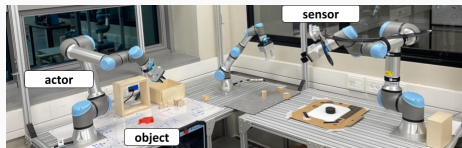
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Formal Models for Sequential Decision-Making

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How to model these settings? What are the states, actions, rewards, and observations? What is the objective?

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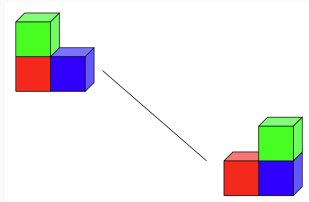
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- Route selection (from Arad to Bucharest)
- Solving 15-puzzle (or Rubik's cube, or ...)
- Selecting and ordering movements of an elevator or a crane
- Production lines control
- Autonomous robots
- Crisis management
- ...

What do all these have in common?

Planning and sequential decision making

- All these problems deal with **action selection** or **control**
- Some notion of problem **state**
- (Often) specification of **initial state** and/or **goal state**
- Legal moves or **actions** that allow transition from states to other states



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What do we need to model ?

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What do we need to model ?

- Sequential decision making
- Uncertainty in action outcomes
- Partial observability

Additional considerations:

The model needs to be compact enough to be solvable but informative enough to be useful.

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Representation Complexity

The **representation complexity** of a model is the amount of information required to specify it.

Formally, for a model class $\mathcal{M}$ and a particular model $M \in \mathcal{M}$, the representation complexity of M is the number of parameters, entries, or bits needed to encode its components, i.e., its states, actions, transitions, observations, rewards, ...

It captures the size of the *input description* of the decision-making problem, as opposed to the time needed to solve it (which we will discuss when considering solution approaches).

Key challenge: model rich enough to be useful vs. simple enough to solve

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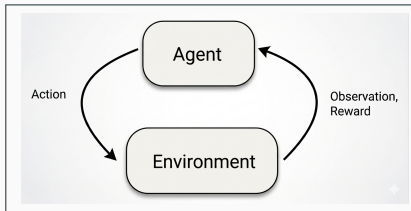
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Modeling the Environment



Is this always a reasonable assumption?

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Different classes of problems

Properties

- **Dynamics:** deterministic, nondeterministic or probabilistic
- **Observability:** full, partial, or none
- **Horizon:** finite or infinite
- ...

Side comment ...

- It is not that deterministic problems are easy
- It is not even clear that they are too far from modeling real-world problems well!

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Properties of the setting: dynamics

Deterministic dynamics

Action + current state **uniquely** determine successor state.

Nondeterministic dynamics

For each action and current state there may be **several possible** successor states.

Probabilistic dynamics

For each action and current state there is a **probability distribution** over possible successor states.

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The Agent and its Decision Making Problem

An agent typically includes one or more of these components:

- **Model:** agent's representation of the environment:
- **Value function:** capture how good is each state and/or action
- **Policy:** agent's behaviour function

We will consider three environment models of increasing complexity:

- Deterministic fully observable environments
- Stochastic fully observable environments
- Stochastic partially observable environments

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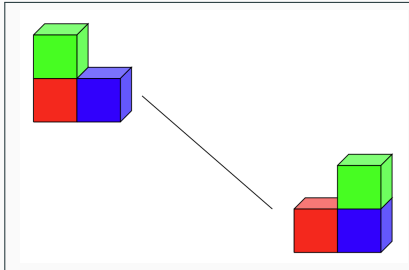
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Deterministic Fully-Observable Domains

Deterministic dynamics example

Pick and Place: Moving objects with a robotic hand:

```
stack(green_block, blue_block)
```



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Propositional STRIPS

Deterministic fully observable domains can be defined using **Propositional STRIPS** (Fikes and Nilsson 1971) where a task is a tuple $P = \langle \mathcal{F}, I, \mathcal{A}, G, \mathcal{C} \rangle$, where

- $\mathcal{F}$ is a set of features (features) and a state s is represented by the features that are true in s
- $I \subseteq \mathcal{F}$ is the initial state
- $G \subseteq \mathcal{F}$ represents the set of goal states, and
- $\mathcal{A}$ is a set of actions.
 - Each action is a triple $a = \langle pre(a), add(a), del(a) \rangle$, which represents the precondition, add, and delete lists respectively, all subsets of $\mathcal{F}$.
 - An action a is applicable in state s if $pre(a) \subseteq s$.
 - If action a is applied in state s , it results in a new state $s' = (s \setminus del(a)) \cup add(a)$.
- $\mathcal{C} : \mathcal{A} \rightarrow \mathbb{R}_0^+$ is a cost function that assigns each action a non-negative cost.

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Example: Pick-and-Place in STRIPS

Goal

$$G = \{ on(green_block, blue_block) \}$$

Relevant features

- $on(x, y)$: block x is on block y
- $ontable(x)$: block x is on the table
- $clear(x)$: nothing is on x
- $holding(x)$: robot holds x
- $handempty$: nothing being held by the gripper

Example action: **stack(x,y)**

$$pre = \{ holding(x), clear(y) \}$$

$$add = \{ on(x, y), clear(x), handempty \}$$

$$del = \{ holding(x), clear(y) \}$$

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Deterministic fully observable environments

The objective is to find a plan $\pi = \langle a_1, \dots, a_n \rangle$: a sequence of actions that brings an agent from the initial state to a goal state.

The cost $c(\pi)$ of a plan π is $\sum_{i=1}^n (C(a_i))$. Often, the objective is to find an optimal solution for P : a plan π^* that minimizes the associated cost.

The literature is rich with different approaches developed to solve the planning problem (Bonet and Geffner 2013): more on this later on in the course.

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Representation Complexity of STRIPS

In a planning problem with feature set $\mathcal{F}$, the number of possible states is $2^{|\mathcal{F}|}$. Thus, the state space grows **exponentially** in the number of features.

If transitions are represented explicitly as a graph over states, then in the worst case there can be $2^{|\mathcal{F}|} \times 2^{|\mathcal{F}|} = 2^{2|\mathcal{F}|}$ possible state-transition pairs.

In contrast, the representation complexity of a STRIPS task is the size of its description:

$$O(|\mathcal{F}| + |I| + |G| + \sum_{a \in \mathcal{A}} (|pre(a)| + |add(a)| + |del(a)|))$$

These represent the total number of features and action literals needed to describe the problem.

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Representation Complexity of STRIPS

In the worst case, if no abstraction is captured in the action description (and all transitions are captured explicitly), the complexity of STRIPS and of an explicit transition function is the same.

However, in STRIPS, there is a one-to-many mapping from action schema to actions (i.e., possible transitions) so it provides a **compact representation** of planning problems.

Therefore a **small description** may induce a **huge state space**.

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Example: Pick-and-Place in STRIPS - representation complexity

In a transition system encoding of **stack(x,y)**, with 3 blocks we have 12 transitions, with 4 blocks we have 84 transitions, and so on...



With STRIPS, we have a single action (for which the representation includes 7 features) regardless of the size of the feature space $|\mathcal{F}|$.

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Uncertain Action Outcomes

From Deterministic to Stochastic Domains

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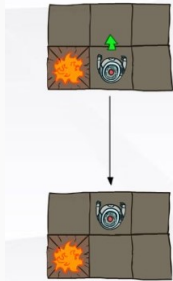
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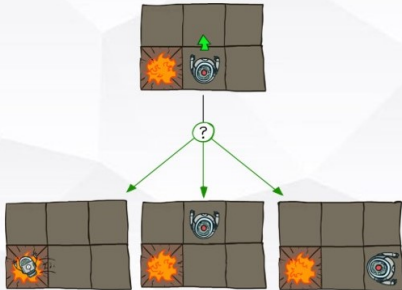
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Deterministic Grid World



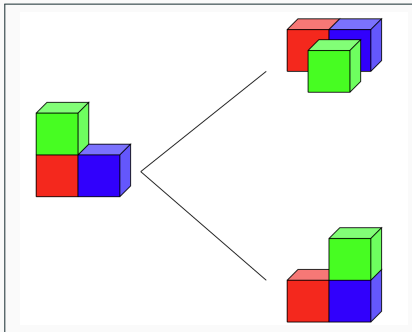
Stochastic Grid World



From IAI

Nondeterministic dynamics example

Moving objects with an **unreliable** robotic hand:
move the green block onto the blue block.



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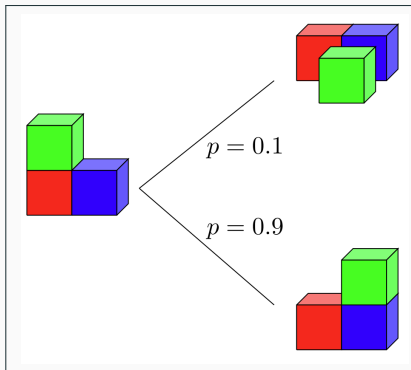
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Probabilistic dynamics example

Moving objects with an **unreliable** robotic hand:
move the green block onto the blue block.



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Stochastic fully observable environments

Stochastic fully observable environments are defined using a **Markov Decision Process (MDP)** $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix
$$\mathcal{P}_{s,s'}^a = \mathcal{P}[S_{t+1} = s' | S_t = s, A_t = a]$$
- $\mathcal{R}$ is a reward function, $\mathcal{R}_s^a = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

The Markov property: “The future is independent of the past given the present”.

Extensions: Infinite and continuous MDPs, partially observable MDPs, undiscounted, average reward MDPs. etc.

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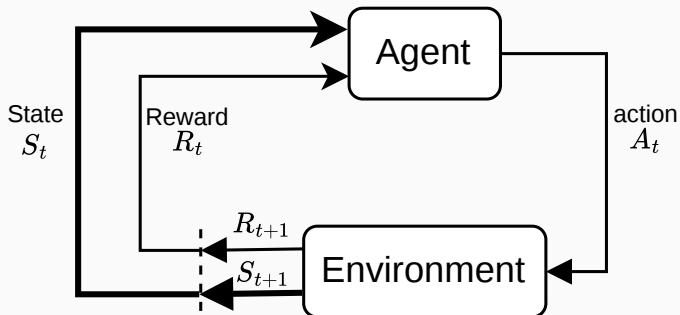
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Stochastic Fully Observable Environments



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A solution to an MDP is a **policy** that prescribes the action to be taken at every state.

- **Deterministic Policy:** a mapping $\pi : \mathcal{S} \rightarrow \mathcal{A}$ from states to actions.
- **Stochastic Policy:** a mapping $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ from state and action pairs to the probability $\pi(a|s)$ of taking action a when in state s .

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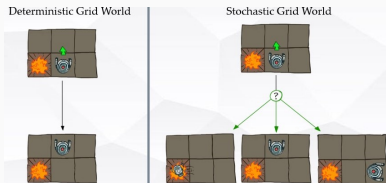
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From Deterministic to Stochastic Domains

There is at least one (deterministic) policy that is better than or equal to all other policies. This is an **Optimal Policy** π^* .



If we know there is an optimal **deterministic** policy for any discounted infinite horizon / finite horizon / stochastic shortest path MDP, why do we need a **stochastic** policy?

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Factored Representation of Markov Decision Process

Typically, the state space $\mathcal{S}$ is represented using feature space $\mathcal{F}$, such that $s \in d^{|\mathcal{F}|}$, where d is the feature dimension.

The MDP is then $\langle \mathcal{F}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

The explicit representation of the state space $\mathcal{S}$ is replaced by a **factored state representation**, where the set of states is described via a set of random variables (features) $\mathcal{F} = f_1, \dots, f_n$, and where each variable f_i takes on values in some finite domain $Dom(f_i)$.

A state is an assignment of a value $f_i \in Dom(f_i)$ for each variable in $\mathcal{F}$.

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Factored Representation of Markov Decision Process

- Factored MDPs allow representing large, structured MDPs compactly.
- Transition dynamics can then be described compactly, e.g. using a **Dynamic Bayesian network (DBN)** (Dean and Kanazawa, 1989). This exploits the fact that the transition of a variable often depends only on a small number.
- Momentary rewards can often also be decomposed as a sum of rewards related to individual variables or small clusters of variables

*Boutilier, Craig, Richard Dearden, and Moisés Goldszmidt. "Stochastic dynamic programming with factored representations." Artificial intelligence 121:1-2 (2000): 49-107 and Guestrin, C., Koller, D., Parr, R., and Venkataraman, S. (2003). Efficient solution algorithms for factored MDPs. Journal of Artificial Intelligence Research, 19, 399-468.

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Factored Representations: Key Idea

Instead of modeling transitions between entire states, we model how individual variables evolve.

This allows us to represent very large systems

- robots
- energy systems
- logistics networks

without enumerating all states.

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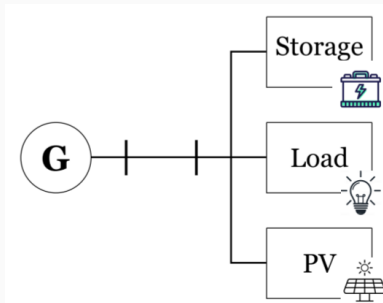
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MDP Example: Battery Management



Consider controlling a battery connected to an electricity grid. The energy from the battery can be used to satisfy local demand or be sold to the grid.

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MDP Example: Battery Management (Flat Representation)

States: The state representation includes all relevant variables: state of charge (soc), price, demand: Example state:

$$s_1 = (\text{soc} = 60\%, \text{price} = \text{high}, \text{demand} = \text{medium})$$

Actions: $\mathcal{A} = \{\text{Charge}, \text{Discharge}, \text{Idle}\}$

Transition model: flat representation must be specified for every triple (s, a, s') .

$$\mathcal{P}(s, a, s') = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a)$$

Reward $\mathcal{R}(s, a, s')$

- profit from selling electricity
- cost of purchasing electricity

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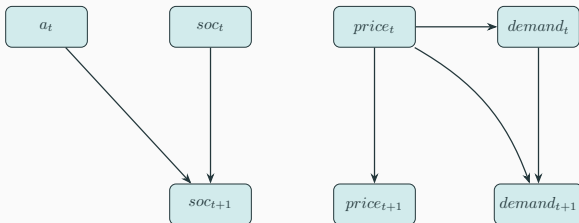
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MDP Example: Battery Management (Factored Representation)

Structured transitions: Instead of specifying $\mathcal{P}(s, a, s')$ for all states, we can encode the dependencies:

- soc_{t+1} depends on (soc_t, a_t)
- $price_{t+1}$ depends on $(price_t)$
- $demand_{t+1}$ depends on $(demand_t, price_t)$

These dependencies can be represented compactly using a **Dynamic Bayesian Network**.



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Representation Complexity: Flat vs Factored MDP

Total number of states

$$|\mathcal{S}| = 11 \times 3 \times 3 = 99$$

With a **flat MDP representation**, for each state-action pair we must specify a distribution over next states: $O(|\mathcal{S}|^2|\mathcal{A}|)$.

With $|\mathcal{A}| = 3$, we have $99^2 \times 3 = 29,403$ transition probabilities.

With a **factored representation**, we only specify transitions for each variable:

- $soc_{t+1} \mid (soc_t, a_t)$
- $price_{t+1} \mid (price_t)$
- $demand_{t+1} \mid (demand_t, price_t)$

Number of parameters is proportional to the number of **local dependencies**.

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In deterministic settings, the state space increased exponentially with the number of features, and the worst-case action space size was induced by the size of the state space

In stochastic settings, transitions must specify a **probability distribution** over successor states. In the worst case, each action may lead to any other state.

If $|\mathcal{S}| = n$ and $|\mathcal{A}| = m$, then an explicit transition model may require specifying $O(mn^2)$ transition probabilities.

In practice, much of modern planning and RL is about coping with **large or continuous state spaces**.

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Partial Observability

Properties of the setting: observability

Full observability

Observations/sensing determine the current world state **uniquely**.

Partial observability

Observations determine current world state **only partially**:
We only know that the current state is one of several possible ones.

No observability

There are **no observations** to narrow down possible current states. However, can use knowledge of **action dynamics** to deduce which states we might be in (a.k.a funneling).

Consequence: If observability is not full, must represent the **knowledge** an agent has.

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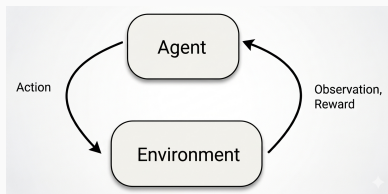
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Agency under partial observability

Recall that the environment-agent interactions give rise to a sequence or trajectory: $s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, r_3, \dots$

From the point of view of the partially informed agent, we have a **history**: $o_0, a_0, r_1, o_1, a_1, r_2, o_2, a_2, r_3, \dots$



The agent needs to maintain its belief based on the current (possibly null) observation.

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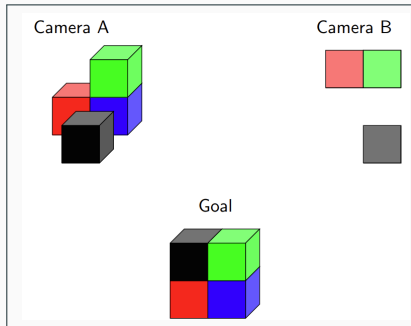
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From IAI



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- In partially observable domains, we may have a **sensor model** represented as a mapping function from what is observed to the actual world state.
- At each time step the agent receives an observation $o_t \in \Omega$ that reflects the current state of the world.
- A **belief** $\beta \in \mathcal{B}$ represents the possible world states (i.e., the states that are deemed possible by the agent).
- Induced by an **observability/ sensor function** $\mathcal{O}_{\mathcal{S},a}^o$ that maps states to observations:
 - non-deterministic $\mathcal{O} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}[\Omega]$
 - probabilistic: $\mathcal{O}_{s,a}^o = \mathcal{P}[O_{t+1} = o | S_t = s, A_t = a]$

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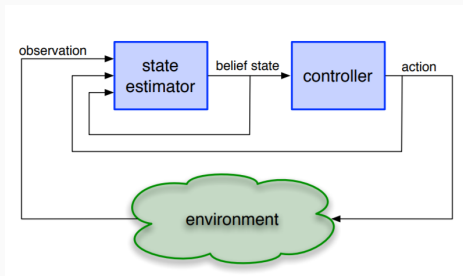
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Beliefs and Belief Tracking

The agent maintains its belief via a **state estimator** - which we will refer to as the process of **belief tracking**.



From Kaelbling, L. P., and T. Lozano-Perez. "Integrated Task and Motion Planning in Belief Space" 2013 https://dspace.mit.edu/bitstream/handle/1721.1/87038/Kaelbling_Integrated%20task.pdf?sequence=1&isAllowed=y

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A belief $\beta \in \mathcal{B}$ can have different representations:

- **Deterministic:** a direct mapping from states to beliefs $\beta = s$ (in which case we simply talk about states)
- **Non-deterministic:** $\beta \subseteq \mathcal{S}$ represents the set of possible world states
- **Stochastic:** the belief is a probability distribution over possible underlying world states. $\beta : \mathcal{S} \rightarrow \mathcal{P}[\mathcal{S}]$ such that $\beta(s)$ represents the probability that s is the actual world state.

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Planning Under Partial Observability with Logical Models

- The true state of the environment is not known, yet partial information about the state is assumed to be available from sensors.
- Uncertainty is represented by sets of possible world states, referred to as **beliefs**.
- In a logical model, a belief $\beta \in \mathcal{B}$ is a subset of states, i.e., $\beta \subseteq \mathcal{S}$

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Planning Under Partial Observability with Logical Models

A *planning under partial observability problem* is a tuple $P = \langle \mathcal{F}, \mathcal{A}, G, \mathcal{O}, I \rangle$ where

- $\mathcal{F}$ is a set of feature symbols,
- $\mathcal{A}$ is a set of deterministic actions,
- G is a set of $\mathcal{F}$ -literals defining the goal condition, and
- $\mathcal{O}$ represents the agent sensor model,
- I is a set of *clauses* over $\mathcal{F}$ -literals (referred to as *facts*) defining the initial situation

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We see the objects, but what is their color?

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Planning Under Partial Observability with Logical Models

A robot must pick up a key that is hidden in one of two drawers.

- **Fluents:**

$$\mathcal{F} = \{InLeft, InRight, OpenedLeft, OpenedRight, HasKey\}$$

- **Actions:**

$$\mathcal{A} = \{openLeft, openRight, takeKeyLeft, takeKeyRight\}$$

where actions are deterministic.

- **Goal:**

$$G = \{HasKey\}$$

- **Initial situation:**

$$I = \left\{ \begin{array}{l} InLeft \vee InRight, \neg(InLeft \wedge InRight), \\ \neg OpenedLeft, \neg OpenedRight, \neg HasKey \end{array} \right\}$$

The agent initially knows the key is in exactly one drawer, but does not know which one.

- **Sensor Model:** After opening a drawer, the sensor reveals whether the key is inside that drawer.

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Initial belief: The agent does not know which drawer contains the key.

		<i>InLeft</i>	<i>InRight</i>	<i>OpenedLeft</i>	<i>OpenedRight</i>	<i>HasKey</i>
$\beta_0 =$	s_1	1	0	0	0	0
	s_2	0	1	0	0	0

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Initial belief: The agent does not know which drawer contains the key.

		<i>InLeft</i>	<i>InRight</i>	<i>OpenedLeft</i>	<i>OpenedRight</i>	<i>HasKey</i>
$\beta_0 =$	s_1	1	0	0	0	0
	s_2	0	1	0	0	0

Action 1: *openLeft*.

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Initial belief: The agent does not know which drawer contains the key.

		<i>InLeft</i>	<i>InRight</i>	<i>OpenedLeft</i>	<i>OpenedRight</i>	<i>HasKey</i>
$\beta_0 =$	s_1	1	0	0	0	0
	s_2	0	1	0	0	0

Action 1: *openLeft*. If after executing the action, the agent receives the observation *key seen*

$$\beta_1 = \{\{InLeft, \neg InRight, OpenedLeft, \neg OpenedRight, \neg HasKey\}\}$$

Plan: *takeKeyLeft*

Otherwise, if it receives the observation: *key not seen*

$$\beta_1 = \{\{\neg InLeft, InRight, OpenedLeft, \neg OpenedRight, \neg HasKey\}\}$$

Plan: *openRight* $\rightarrow$ *takeKeyRight*

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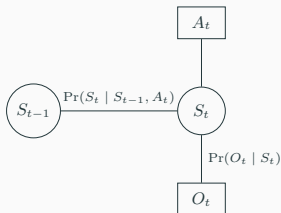
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Probabilistic Belief

A probabilistic sensor model is defined by the probability $\mathcal{O}_{s,a}^o = \mathcal{P}[O_{t+1} = o | S_t = s, A_t = a]$ of observing o when action a is performed at state s .



In probabilistic models, a **belief** represents the possible world states as a probability distribution, i.e., $\beta : \mathcal{S} \rightarrow \mathcal{P}[\mathcal{S}]$ such that $\beta(s)$ represents the probability that s is the actual world state.

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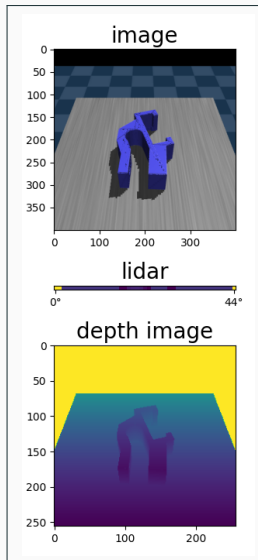
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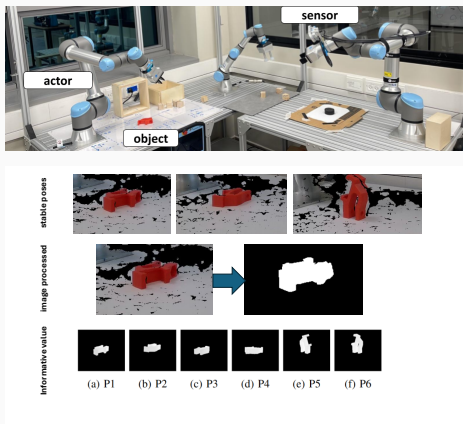
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$$\beta^{o,q}(p) = \frac{\hat{P}(o|p, q) \beta(p)}{\int_{p' \in \mathcal{P}} \hat{P}(o|p', q) \beta(p') dp'} \quad (1)$$

where $\beta(p)$ is the estimated probability that p is the pose prior to considering observation o .

Partially Observable Markov Decision Process (POMDP)

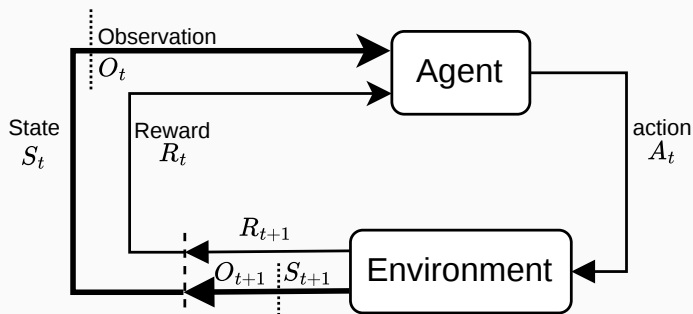
A **Partially Observable Markov Decision Process**(POMDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$ where

- $\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}$ and γ are as for an MDP.
- Ω is a set of observations (observation tokens),
- $\mathcal{O}$ is a sensor function specifying the conditional observation probabilities $\mathcal{O}_{s,a}^o = \mathcal{P}[O_{t+1} = o | S_t = s, A_t = a]$ of receiving observation token $o \in \Omega$ in state s after applying action a ¹.
- β_0 the initial belief: a probability distribution over the states such that $\beta_0(s)$ stands for the probability of s being the true initial state.

Markov property still holds

¹alternatively: $\mathcal{O}_s^o = \mathcal{P}[O_t = o | S_t = s]$

Stochastic Partially Observable Environments



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A solution to a POMDP is a **policy** that prescribes the action to be taken at every belief.

- **Deterministic Policy:** a mapping $\pi : \mathcal{B} \rightarrow \mathcal{A}$ from beliefs to actions.
- **Stochastic Policy:** a mapping $\pi : \mathcal{B} \times \mathcal{A} \rightarrow [0, 1]$ from belief and action pairs to the probability $\pi(a|\beta)$ of taking action a when in belief β .

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Representation Complexity

- A POMDP extends an MDP by adding an observation space Ω and an observation model $\mathcal{O}$.
- If $|\mathcal{S}| = n$, $|\mathcal{A}| = m$, and $|\Omega| = k$, then an explicit representation may require:
 - $O(mn^2)$ transition probabilities for $\mathcal{P}$
 - $O(mnk)$ observation probabilities for $\mathcal{O}$
- However, the main increase in complexity comes from the fact that policies are defined over **beliefs**, not states.
- Since beliefs are distributions over $\mathcal{S}$, the belief space is continuous even when $\mathcal{S}$ is finite.

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World Models

World Models are internal representations of an environment that an AI agent builds to understand physical dynamics, predict future states, and plan actions.

Unlike Large Language Models (LLMs) that predict the next token in a sequence of text, world models aim to "imagine" or simulate the physical consequences of actions.

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Foundation Models as World Models

- A **world model** is used to predict how the environment evolves and what observations or outcomes may follow an action. They include:
 - transition models
 - observation models
 - reward models
- Traditionally, world models were hand-designed.
- Increasingly, these models are **learned from data** using large neural networks and foundation models.

The development of world models is a key stepping stone toward Artificial General Intelligence (AGI)

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- **Predictive Reasoning:** It takes the current state of the world and a potential action to predict the next state (e.g., "If I push this glass, it will fall and break").
- **Physical Intuition:** These models learn laws like gravity, momentum, and object permanence through observation rather than programmed equations.
- **Safe Exploration:** AI agents can "dream" or simulate thousands of scenarios internally, learning from mistakes without damaging hardware or risking safety in the real world.

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Foundation Models as World Models

Promise:

- richer generalization across tasks and environments
- less manual engineering of the model

Challenges:

- predictions may be approximate, biased, or hard to verify
- planning with learned world models requires handling uncertainty and model errors

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Common Architecture of a World Model

A **world model** learns a predictive representation of the environment that allows the agent to:

- encode observations into a **latent state space**
- predict how the world evolves via a **dynamics model**
- predict **rewards** and optionally **future observations**
- **plan or learn policies** by simulating trajectories inside the model

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Major Players and Models (March 2026)

- Genie 3 (Google DeepMind) - Real-time interactive 3D environments and gaming <https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models>
- Cosmos (NVIDIA) - Physical AI and industrial digital twins
- GWM-1 (Runway) - Controllable video generation and robotic manipulation <https://runwayml.com/research/introducing-runway-gwm-1>
- Marble (World Labs) - Multimodal 3D world generation from text or images <https://runwayml.com/research/introducing-runway-gwm-1>
- JEPA Research (AMI Labs) - Learning world representations from sensory data <https://futurumgroup.com/insights/yann-lecuns-ami-raises-1bn-seed-round-is-the-world-mod>

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- Robotics: Allowing robots to practice complex tasks (like folding laundry or navigating a factory) in simulation before deployment.
- Autonomous Vehicles: [Waymo]use world models to simulate rare "corner cases" in traffic to train self-driving stacks more safely.
- Gaming: Generating infinite, interactive environments where every object reacts realistically to player input.
- Industrial Digital Twins: Creating virtual replicas of warehouses or factories to test new layouts or workflows without disruption.

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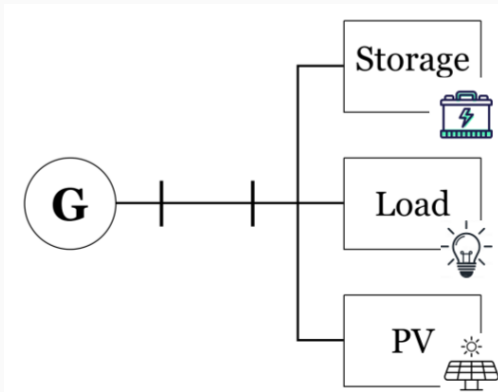
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What is the best model?

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Complex Decision Making in Robotics

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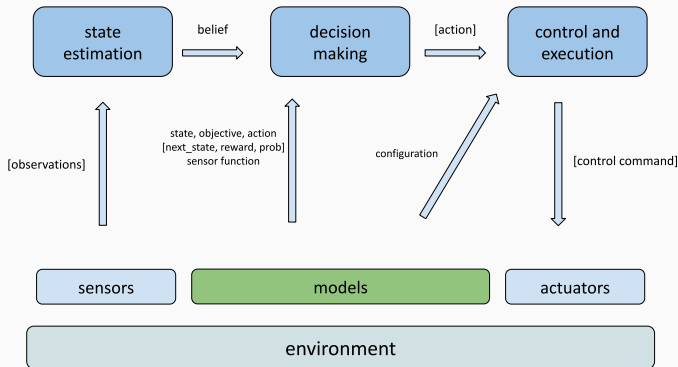
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- Logical vs. probabilistic models for partial observability
- Do we need to model the sensor function (and use a POMDP) or can we use an MDP ?



Fig. 1: Our method learns visuomotor policies that directly

End-to-End Training of Deep Visuomotor Policies Levine et al. 2016

<https://www.jmlr.org/papers/volume17/15-522/15-522.pdf>

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Summary

- Models for decision making in AI for single and multi-agent settings
- Limitations and benefits of each model

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